IU Internationale Hochschule

SMS Spam Detection

NLP Project Report

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Abstract

This project examines whether a model detects more spam when trained on more SMS messages. The messages originate from the SMS Spam Collection. The texts have been converted into numbers with TF-IDF and classified with Multinomial Naive Bayes. Training took place with 500, 2,000 and lastly with 4,128 SMS. The same 1,032 test SMS were used for all three runs.

The proportion of the detected spam SMS increased from 27.3% to 67.2% and then to 74.2%. The biggest improvement was made between 500 and 2,000 text messages. With the biggest training set the model detected 95 out of 128 spam SMS. One normal text message was incorrectly marked as spam.

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1 Introduction

A spam filter should detect unwanted SMS. There are two possible errors: A spam SMS could reach the normal message inbox, or a normal (not spam) message could incorrectly be classified as spam and be filtered out. The second scenario is an especially big problem if it is an important message. For the assessment of the filter, it is therefore important which errors are made and how often.

This project examines how the amount of training data affects the quality of the spam detection. The research question is: Does the model detect a larger proportion of the spam SMS if it is trained with more messages? This proportion is called spam recall. For the comparison, three models were trained with the same method, but with different amounts of training data. All the models were tested on the same test messages.

The report first describes the dataset and the project workflow. It then explains how the messages were prepared and how the model was trained. Finally, it presents the results and discusses the limitations.

1.1 Related Work

Almeida et al. (2011) introduced the SMS Spam Collection and compared several methods for detecting SMS spam. In their approach a support vector machine delivered the best results. This project pursues a different aim: it aims to see how the amount of training data affects the quality of the detection.

For this purpose, Multinomial Naive Bayes was chosen. The method can classify texts based on word features and can be implemented using the scikit-learn library. Scikit-learn also describes the use of TF-IDF values with the same model (scikit-learn developers, n.d.). In all three runs the same method was used with the same settings.

2 Project Planning and Dataset

2.1 Project Workflow and Tools

The project was structured into three steps. First, the SMS were imported, verified and split. After that, the texts were prepared and the models were trained. Finally, the results were compared and the model with the largest training set was tested with two SMS messages the author had composed herself. The order is important because no information from the test data should be used during training.

The implementation took place in Google Colab. The messages were loaded using pandas and processed as a table. NLTK was used to stem the English words. Scikit-learn was used for splitting the data, the conversion of the texts to numbers, the training and the evaluation. The notebook contains the individual steps and the corresponding outputs. Therefore, the sequence can be reproduced in the same order.

Notebook in GitHub: https://github.com/KyraPal/nlp-task2-sms-spam-detection

2.2 Dataset and Split

The SMS Spam Collection includes 5,574 English messages (Almeida & Hidalgo, 2011). Every SMS has a text and a label. The label ham stands for a normal SMS, spam for an unwanted message. The original dataset includes 4,827 normal messages and 747 spam messages. Before splitting, spaces at the beginning and end of the texts were removed. After that, 414 entries with the same message text were removed. This ensured that no identical text could appear in both the training and the test sets. Similar messages with minor variations could still remain in the dataset. After cleaning, 5,160 SMS messages remained. About 80% of the cleaned-up data was used for training and 20% for testing. When making the split, care was taken to ensure that the proportion of spam remained roughly the same in both groups. As shown in Table 1, this is about 12%. The test set consisted of 128 spam SMS and 904 normal SMS. This unequal distribution is important for the later evaluation.

Table 1: Distribution of normal and spam SMS

Dataset	normal SMS	Spam SMS	total
Original dataset	4,827	747	5,574
Dataset after removing duplicates	4,518	642	5,160
Small training set	438	62	500
Medium training set	1,751	249	2,000
Full training set	3,614	514	4,128
Test set	904	128	1,032

3 Method

3.1 Text Preprocessing

Initially, all letters were converted to lowercase. Because of this, for example, FREE and free are treated the same. After that, dots, commas and other common punctuation marks were replaced with spaces. This makes it possible to separate the words from each other. Common English words like “the” and “and” were removed, so the model can focus on the remaining words. After that, the words were shortened using the Porter stemmer. For example, “winning” becomes “win”. This step is called stemming and helps to group similar word forms. After the preparation, “win”, “free” and “ticket” remained. The same preparation was applied to the training data, the test data and the new example SMS.

3.2 Transformation of texts using TF-IDF

To ensure that the model can work with the SMS, the words were converted into numbers using TF-IDF. A word tends to be given more weight if it appears frequently within a single text message but occurs in only a few of the messages in the training set overall. A word which occurs in more messages therefore gets less weight than a comparatively rare word. Every SMS results in a series of numbers that match the words in the word list. For every run, the list of words and the weights were determined solely from the respective training messages. After that, the test SMS messages were transformed using this exact word list. A word that appears only in the test set and is missing from the word list cannot be used by the model in this run. Therefore, the test can show how well the model can process messages it has not seen during training.

The values in the notebook also serve as a check on the data processing. In every run, the test data consists of 1,032 rows, meaning one row per test SMS. The number of columns changes with the list of words. In each run, the training and test data have the same number of columns. This is because each column in both sets represents the same word feature.

3.3 Training with Naive Bayes

During training, the model receives the TF-IDF values and the correct label of every training SMS. Using this, the model learns which word features tend to appear in spam and normal messages. Furthermore, it takes into account how often both groups appear in the training data. For a new message, the model compares the two possible labels and chooses the more likely classification (scikit-learn developers, n.d.).

A promise of a prize, for example, may contain words that the model has encountered more frequently in spam. A single word like “free” is not automatically enough to label an SMS as spam. The model also considers all other detected words. It reviews every single word feature

without understanding the meaning of the sentence as a human would. This is a limitation of the chosen approach.

Training took place with 500, 2,000 and 4,128 SMS. The 500 messages are part of the intermediate dataset, and the 2,000 messages are part of the total training dataset. This ensures that the SMS messages used previously are retained when switching to a larger quantity. The proportion of spam stayed approximately the same. The selection is reproducible due to the fixed random seed value of 7. The 1,032 test SMS messages were not used for training in any of the three runs.

The smallest training dataset included 62 spam SMS. In the medium-sized set, there were 249, and in the largest set, 514. This kept the proportion similar while increasing the number of examples for both groups. In this way, the project examined the impact of additional training messages while maintaining a comparable ratio of spam to normal SMS.

3.4 Evaluation of the predictions

For the evaluation, the predictions were compared to the existing labels of the test data. Accuracy indicates the proportion of all SMS messages that were classified correctly. The spam recall describes only the actual spam messages and indicates what proportion of them was detected correctly. Because normal messages occur more often in the dataset, the accuracy can stay high even if a lot of spam messages remain undetected. Spam precision, on the other hand, considers all messages that the model has marked as spam. It shows what proportion of them were actually spam. The spam F1-score combines precision and recall. This makes it possible to assess not only the proportion of spam messages detected but also the reliability of the spam labels.

4 Results and Reflection

4.1 Comparison of the Three Training Sets

Table 2 shows the results for the same 1,032 test SMS. With 500 training messages, the model detected 35 out of 128 spam SMS. With 2,000 training messages it detected 86. This results in 51 additionally detected spam messages. When switching to 4,128 training messages, another nine were added. The biggest improvement was made between the small and the medium training set.

Table 2: Results on the same 1,032 test messages

Training messages	Accuracy	Spam-Precision	Spam-Recall	Spam-F1	Spam messages detected
500	91.0%	100.0%	27.3%	42.9%	35 of 128
2,000	95.9%	100.0%	67.2%	80.4%	86 of 128
4,128	96.7%	99.0%	74.2%	84.8%	95 of 128

In the first two runs, the spam precision was 100%. All messages detected as spam were indeed real spam messages. Nevertheless, the model with the smallest training set missed 93 of the 128 spam messages. Here, the high precision means that the labels were reliable. How much spam was missed is shown by the recall.

Across all three runs, the accuracy increased by 5.7 percentage points. For spam recall, this increase was 46.9 percentage points. The improvement is more pronounced in terms of recall because accuracy also factors in the many normal messages. All of these were correctly classified already in the first run. Consequently, there was more room for improvement when it came to spam detection.

Accuracy can also be better put into context through a simple comparison. A model that classifies every message as normal would already reach an accuracy of 87.6% because it would classify 904 out of 1,032 messages correctly. However, it would not detect a single spam SMS. This is why the initial accuracy of 91% only seems high at first glance. The spam recall of 27.3% shows that the filter detects only around a quarter of the unwanted messages.

4.2 Errors of the largest model

Figure 1 shows the predictions of the model trained on 4,128 SMS messages. The rows represent the actual labels, while the columns represent the predictions. The model correctly classified 903 normal SMS messages and detected 95 spam messages. It missed 33 spam messages and incorrectly labeled one normal message as spam.

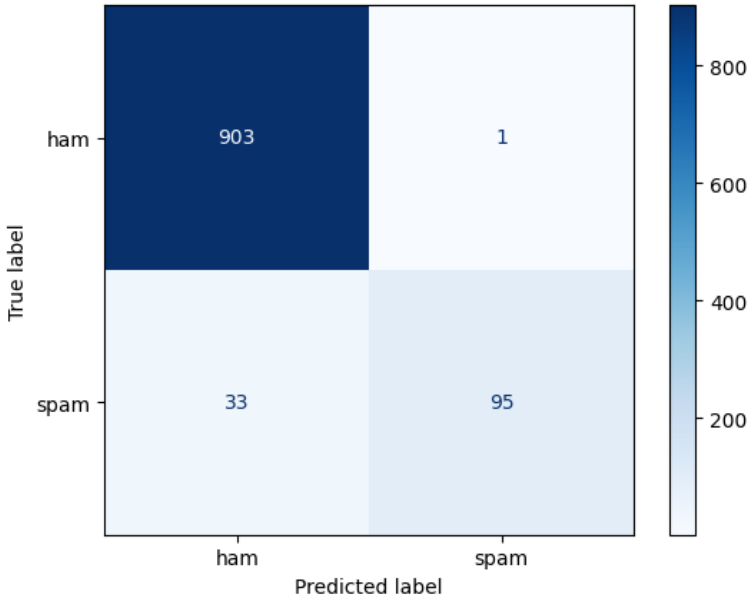


Figure 1: Confusion matrix for the model with 4,128 training SMS messages

One of the spam text messages that went undetected contained a request to call a number. During the preparation of the text, the words “please” and "call" were removed because they were on the stop-word list. Consequently, the model could not take these words into account when making its decision. However, this alone does not explain why it classified the text message as a normal message.

4.3 Test with two new SMS

Finally, the model with the largest training set was tested with two self-composed messages. The question “Are we still meeting at six tonight?” was labeled as a normal message. The promise of a cash prize “Congratulations! Claim your free cash prize now!” was detected as spam. Both texts were prepared in the same way as the test SMS and converted using the existing word list. The model was not trained again.

Both predictions show how a new SMS can be classified in the notebook. The messages are deliberately easy to distinguish. Therefore, it is not possible to say how reliable the filter would be in everyday use. In this regard, the comparison based on the 1,032 test SMS messages remains more meaningful.

4.4 Limitations of the comparison and potential application

A possible explanation for the improvement is that the model with the larger training set has seen more typical spellings and words. The word list grew from 1,823 features with 500 training SMS to 4,147 features with 2,000 SMS and to 6,423 in the largest training dataset. At the same time, the model had more examples available for assigning the words to the two classes. The experiment does not separate these two influences. Therefore, it cannot be concluded that the

larger word list alone explains why more spam was detected.
An advantage of the setup is that all models were evaluated on the same set of test messages. However, the results came from a single split. With a different split, other spam messages could end up in the test set, and the values might change. A repetition with different random splits would show whether the advantage of the larger training set remains across different splits.

The dataset contains English SMS. The results therefore cannot be applied to messages in other languages without further testing. Current spam SMS messages may also contain different terms or spellings. For practical use, a separate spam folder would be useful so that any message incorrectly filtered out could be retrieved. Whether the filter would be reliable enough for this purpose would need to be tested using current messages.

For a test like this, it would be reasonable to analyze the incorrect decisions separately. If a normal text message ends up in the spam folder, it must be possible to find it there. If a spam text message remains in the normal inbox, the consequences depend on its content. An unwanted promotional message should be evaluated differently from a message that requests a suspicious callback. The evaluation of the filter also depends on which types of errors carry the most weight in each use case.

5 Conclusion

The research question for this project can be answered as follows: with more training SMS, the model detected a larger proportion of spam messages. Spam recall increased from 27.3% with 500 training messages to 74.2% with 4,128. The largest improvement occurred between 500 and 2,000 SMS. Even the model with the largest training set still missed 33 spam messages and incorrectly filtered out one normal message.

The comparison also shows why multiple metrics are necessary. Accuracy describes the overall result, spam recall shows how much spam was missed, and spam precision helps in evaluating cases where messages are incorrectly flagged as spam. As a next step, it would be useful to repeat the comparison using different data splits. Testing with more recent messages would also show whether the model is also capable of detecting other types of SMS spam.

One takeaway for future AI projects is to check what information may be lost when preparing text for a model. In this project, reviewing a missed spam message showed that the stop-word list had removed “call”, even though the request to call was an important part of the message.

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